REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ONE WORLD, BISHOP ALPHA AND KINDOROKO SECONARY SCHOOLS
FORM THREE JOINT TERMINAL EXAMINATION

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 3:00 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on every page of your answer booklet(s).

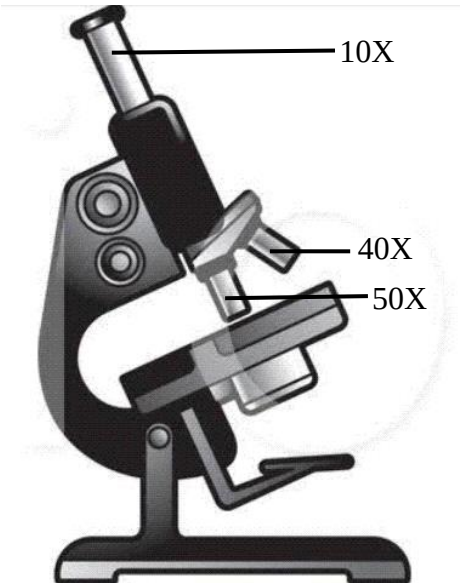
FORM THREE TERMINAL 2025

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer booklet(s) provided.

(i) Study the diagram below that shows the light (compound) microscope. Which of the following is the correct magnification of an organism being observed?



A 40X
B 50X
C 200X
D 400X
E 500X

(ii) When deoxygenated blood exits the heart towards being oxygenated, it cannot flow back into the heart. This is enhanced by the presence of:

A Pulmonary semilunar valve
B Aortic semilunar valve
C Mitral valve
D Tricuspid valve
E Bicuspid valve

(iii) Animals are not plants. This is true because:

A Unlike plants, animals do not locomote
B Both animals and plants cells have nucleus
C Animal cells stores glycogen, but not plant cells
D Both animal and plant cells have cell walls
E Plants respire aerobically, but not animals.

(iv) The main function of swim bladder of a fish is:

A To alter the body density of a fish
B To store water when a fish swims fast
C To store urinary wastes
D To store unfertilized eggs
E To regulate the viscosity of a fish

(v) A certain food sample was tested in the laboratory. Three drops of Iodine solution were added to a food sample solution in a clean test tube. A blue black colouration was observed. What is the nutrient contained in the food sample, and where is it digested?

A Protein; digested in the stomach
B Starch; digested in the duodenum
C Lipids; digested in the duodenum
D Reducing sugar; digested in the mouth
E Non-reducing sugar; not digested

(vi) Lilian got infected of a certain disease and recovered after a month. Later, her body became resistant to that particular disease such that she cannot be infected of it in future. This situation is clearly defined as:

A Natural passive immunity
B Artificial passive immunity
C Natural active immunity
D Artificial active immunity
E Natural artificial immunity

(vii) When *Mimosa pudica* plants are touched by finger or any other object, they folds their leaves. This is described as:

A Thermonasty
B Thigmonasty
C Photonasty
D Chemonasty
E Geotropism

(viii) Which of the following is the correct order of the air path towards the respiratory structure?

A Trachea – Bronchus – Bronchiole – Alveolus
B Alveolus – Trachea – Bronchiole – Bronchus
C Alveolus – Bronchiole – Bronchus – Trachea
D Bronchus – Bronchiole – Trachea – Alveolus
E Bronchiole – Alveolus – Trachea – Bronchus

(ix) Which of the following is not true about pepsin and trypsin?

A They both act best in acidic medium
B They are both digestive enzymes
C They are both protein digestive enzymes
D They are both produced in inactive forms
E They are both produced in the stomach

(x) Which of the following organism is a protoctists?

A Earthworm
B Tsetse fly
C Paramecium
D Liver fluke
E Vibrio cholerae

2. Match each item in **List A** with the correct response in **List B** by writing its letter below the number of the corresponding item in the answer booklet(s) provided.

LIST A		**LIST B**	
(i)	Bi-concave cells in shape which lack nuclei	A	Leukocytes
(ii)	Irregular shaped cells which respond to infections	B	Erythrocytes
(iii)	Fragment of cells produced in the bone marrow	C	Thrombocytes
(iv)	A fluid tissue consisting of white and red cells, platelets and plasma	D	Thrombin
		E	Group O
(v)	The fluid part of the blood	F	Plasma
(vi)	Commonly known as the universal recipient	G	Capillaries
		H	Group AB
		I	Blood

SECTION B (54 Marks)

Answer **all** questions in this section

3. Suppose you are asked to equip the newly constructed biology laboratory with the most important apparatuses, which apparatuses will you consider and why? Provide six points.

4. (a) Which three features do animal, plant, fungi and amoeba cells have in common?
   (b) With examples, explain four examples of poor waste disposing that you have witnessed in your community.

5. (a) Construct a feeding relationship that accommodates all of the following organisms:
   Deer, Fox, Carrot, Sheep, Toadstool, Wild dog, Grass and Rabbit
   (b) Analyze the reason for why excretion is very important in living things.

6. (a) State two features which help to reduce friction between bones at the movable joints.
   (b) With examples, explain three types of muscles found in the human body.

7. In six points, show how the situation would be if there were no angiosperms in the environment.

8. (a) Distinguish simple from conditioned reflex actions.
   (b) Identify the name of the neuron that carries electrical impulse from the control center to the effector.
   (c) With the aid of the diagram, describe the main parts of the neuron identified in (b) above.

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. Anaerobic respiration in fungi and plants is referred to as fermentation. How is this process significant in today's daily life of people? Provide six points.

10. Explain six methods used to keep fish, vegetables, meat and other raw foods for a long time so that they can be used in future.

11. Sofia was invited by form one students to teach them about HIV/AIDS. Help her to prepare a speech on three transmission ways and three preventive measures against this disaster.

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